

KEERTHI YALAMANCHILI

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Education

Master's in data science, GPA (3.5/4)

August 2023 – May 2025

Indiana University Bloomington, Indiana, USA

Relevant Courses: Data Mining, Advanced Database Concepts, Statistics, Applied Algorithms, Data analytics for research, Cloud Based Analytics, AWS, Data Visualization, Advanced Database Technologies, Visual Design, Social Media Informatics

Bachelors in Electronics and Communication Engineering, GPA (3.64/4)

June 2017 – July 2021

Anurag Group of Institutions, Hyderabad, India

Relevant Courses: Core Java, Object Oriented Programming, Cloud Computing and IoT, Big Data Analytics, Probability Theory and stochastic process, Computer Networks

Experience

Indiana University Bloomington | DATA SUPERVISOR (Inventory & Operations)

August 2024 - April 2025

- Analyzed and SQL queried weekly/monthly inventory data for 50+ food and supply items using Excel, identifying usage patterns to optimize ordering and minimize waste by 10%.
- Developed and maintained SQL-driven Excel and Power BI dashboards to visualize stock levels, KPIs, and seasonal trends, improving ordering accuracy by 15% and reducing stockouts by 10%.
- Investigated 100+ service billing discrepancies per month, liaising with stakeholders to ensure contract compliance and resolve open invoices.

Indiana University Bloomington | DATA ANALYST INTERN (RESEARCH)

May 2024 - August 2024

- Collaborated with faculty mentor using Python and SQL to process 200+ IPO records (2016–2018), accelerating feature engineering by 50% and improving layoff prediction accuracy.
- Collected and parsed IPO prospectuses using Python, employing web scraping (BeautifulSoup, Selenium) and public API endpoints to automate large-scale data extraction.
- Cleaned and structured 100,000+ rows of text and numeric IPO data for downstream analysis with Pandas and NumPy, improving data quality and consistency to support modeling.
- Implemented reproducible data ingestion pipelines and documented ETL scripts/schema, automating data integration to ensure consistency and transparency in the IPO layoff prediction project.
- Conducted exploratory data analysis (EDA) to identify trends and red flags related to firm performance and risk factors associated with layoffs.
- Formulated summary reports and data visualizations using Matplotlib and Seaborn to support hypothesis formation and present findings to the research team.
- Documented Git-backed runbook covering ETL scripts, API endpoints, and data schemas, reducing onboarding errors by 40% and ensuring full pipeline reproducibility.

SOCIETE GENERALE | SOFTWARE ENGINEER

September 2021 - July 2023

- Designed and enhanced features for an onboarding client tool using Java 8 and Spring Boot, enhancing performance and code maintainability across REST APIs and Spring Batch jobs.
- Engineered reusable analytics dashboard components for internal platforms, integrating with backend microservices and SQL sources to automate data processing, streamline pipelines, and boost user efficiency.
- Refactored GDPR batch data pipeline by decoupling core business logic into a dedicated microservice, boosting automation, testability, CI/CD deployment speed by 40%, scalability, and data reproducibility.
- Implemented event-driven pipelines using Apache Kafka to trigger Spring Batch workflows, ensuring reliable cross-system data propagation, real-time processing alerts, and improved pipeline monitoring for compliance.
- Containerized Spring Boot microservices and batch jobs with Docker, establishing consistent development and production environments, reducing configuration drift by 30%, enhancing pipeline stability.
- Deployed microservices to Kubernetes (K8s) and defined Helm charts for batch workloads, streamlining deployment, monitoring, and scaling.
- Engineered Airflow-integrated Spring Batch ETL pipelines to automate secure file transfers and data workflows, significantly reducing manual effort and enabling timely analytics and reporting dashboards.
- Enhanced backend APIs to deliver SQL-enriched datasets for automated ETL pipelines and self-service reporting dashboards, boosting data pipeline efficiency and reporting accuracy across analytics teams.

- Migrated Spring Batch environments from JDK 1.8.0_25 to 1.8.0_192, addressing security vulnerabilities and improving runtime stability.
- Debugged Java path issues causing dropdown errors in batch UI, implementing alternate routing to restore 99% UI consistency and ensure data pipeline stability.
- Mitigated critical SonarQube vulnerabilities (SQL injections, log injections, Log4j issues), improving security posture and code quality.
- Led integration of Prometheus and Grafana dashboards to automate application and batch job pipeline monitoring, enabling real-time observability, proactive alerting, and 30% faster resolution.
- Actively contributed to Craft practices and Agile ceremonies, resolving legacy bugs and reducing tech debt by 30% over multiple sprints.

Skills

Programming: Python, C, C++, Java, SQL, OOP (Object Oriented Programming), REST API, R, SAP

Messaging Services: Kafka, RabbitMQ

Web Components: HTML, CSS, JavaScript, Latex, Springboot, Web Scraping, Spring MVC, React

Cloud & DevOps: AWS (EC2, RDS, S3, IAM, Auto Scaling, Load Balancer, CDK), Airflow, Secrets Manager

Machine Learning & AI: Deep Learning, NLP, Computer Vision, Data Science, LLMs

Tools and platforms Matlab, Git, Pgadmin, SonarQube, JIRA, Jenkins, NI Labview, MySQL, IntelliJ, Power BI, Kubernetes, Docker, SAS

Data Engineering: ETL Pipelines, Data Modeling

Data Visualization: Power BI, Tableau, Matplotlib, Looker, KPI Analysis, Dashboard Development, BI Reporting

Certifications: Cloud certification in **AZ-900**, MATLAB Machine Learning - Onramp, **SQL** Associate certificate

Projects

AWS Capstone: Secure and Scalable Web Infrastructure:

September 2024 - Nov2024

- Architected a secure, scalable AWS cloud infrastructure using Amazon EC2, Amazon RDS, IAM, and an Application Load Balancer (ALB), ensuring high performance and compliance with AWS security best practices.
- Configured AWS Bastion Host and NAT Gateway with an ALB for encrypted SSH access and traffic routing, eliminating 100% of direct server exposure and ensuring audit-ready compliance.
- Centralized 100% of application credentials in AWS Secrets Manager and SSM Parameter Store, encrypting all secrets, removing plaintext storage, and enabling full audit logging for data protection.

AWS Project: High Availability and Scalability

Sep 2024 - Nov2024

- Built a highly available multi-tier AWS architecture across multiple Availability Zones using Amazon EC2 (web servers) and Amazon RDS (database), incorporating Auto Scaling Groups and an Elastic Load Balancer to achieve ~99.9% uptime and fault-tolerant operations.
- Leveraged AWS Auto Scaling and Elastic Load Balancing to dynamically scale resources based on demand, analyzing performance metrics to fine-tune scaling policies and improve application response times by 30% during peak traffic.
- Performed cost analysis using the AWS Pricing Calculator and usage data to optimize resource allocation, reducing projected AWS costs by 20% while maintaining high performance and scalability requirements.

Real Estate Explorer: The Analysis of Real Estate Listings on Zillow Via Web Scraping:

Jan 2024 - Feb 2024

- Automated scraping of **10,000+** Zillow listings with Python (BeautifulSoup, requests, API), structuring pricing, location, and feature data for rapid analysis and decision support.
- Cleaned and transformed **100,000+** HTML-derived records using Pandas within an automated ETL pipeline, reducing data preparation time by **60%** for exploratory analysis.
- Launched interactive Matplotlib and Seaborn dashboards illustrating regional property value trends and feature correlations, empowering **15+** investors with actionable market insights.

Enhancing Online Payment Fraud Detection:

Nov 2023 - Dec 2023

- Performed data cleaning and exploratory analysis on 1M+ online transactions using Python (pandas, matplotlib, seaborn), identifying key fraud indicators through univariate and bivariate analysis.
- Trained and tuned sci-kit-learn classification models (AdaBoost, Gradient Boosting) on imbalanced fraud datasets using Python, boosting recall by 20% and accuracy to 93%.
- Initiated interactive Python dashboard and REST API for real-time fraud scoring, deploying a scikit-learn model to classify transactions live and reducing investigation response time by 50%.